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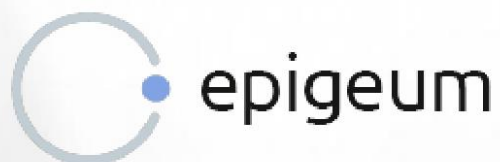
Daniel Tijesu Ohifeme

FOR COMPLETING THE PROGRAMME

Academic Integrity: For Students, Second Edition

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National College of Ireland

Project Submission Sheet

Student Names and ID:Henrique Carlos da Silva Ricardo (24303186)
.....Daniel Tijesu Ohifeme (24220370).....

Programme:Msc Data Analytics..... **Year:** 2026

Module:Business Intelligence and Business Analytics.....

Lecturer:Giovanni Estrada.....

Submission Due Date:24/04/2026.....

Project Title: Supply Chain Crisis and Customer Retention for Nike Ireland

Word Count: 2270.....

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Signature:



Date: 24/04/2026

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1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**
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AI Acknowledgement Supplement

- I. BUSINESS INTELLIGENCE AND BUSINESS ANALYTICS
- II. Supply Chain Crisis and Customer Retention for Nike Ireland

Your Name/Student Number	Course	Date
Henrique Carlos da Silva Ricardo (24303186)	MSc Data Analytics	24/04/2026
Daniel Tijesu Ohifeme (24220370)	MSc Data Analytics	24/04/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

III. AI ACKNOWLEDGMENT

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Not Applicable		

IV. DESCRIPTION OF AI USAGE

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

V. EVIDENCE OF AI USAGE

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

VI. ADDITIONAL EVIDENCE:

[Place evidence here]

ADDITIONAL EVIDENCE:

Supply Chain Crisis and Customer Retention for Nike Ireland

Henrique Carlos da Silva Ricardo- 24303186

Daniel Ohifeme Tijesu - 24220370

24/04/2026

Video Link: <https://youtu.be/A3FGSyJKJNc>

1. Project specification

a) Background information & business goals

Nike is a global leader in performance footwear, clothing and sportswear, they operate in many countries and Ireland included. Although, Nike Ireland, operates in a complex and fast-growing retail environment, where customer experience is based on speed, reliability and service delivery. As a company that relies on a network of suppliers, distribution processes and multichannel sales, customer satisfaction and brand perception gets the most affected when the operations are not being efficient. Despite the demonstration of a strong commercial performance, exceeding €1.5 million during 2024 and 2025, Nike Ireland has experienced a decline in customer satisfaction levels, due to service quality, mostly late deliveries, and it impacts the business, the brand and shows some operational issues. The management biggest concern is the inability to use data to deliver insights into timely and effective actions. As Davenport [1], organisations struggle to analyse operational data resulting in a gap between insights generation and business execution, this gap is evident in Nike's current process, where KPIs such as late delivery rates and stock availability are monitored, but not actively used to improve performance and lead to decision makers. To tackle this problem, the main goal is to develop and implement a Business Intelligence and Customer Relationship Management (CRM) integrated solution that allows the company to operate in a proactive and data driven manner instead of the current reactive manner. The business goals are to reduce late deliveries, improve customer satisfaction, and enhance overall operational efficiency between analytics and business process.

I. EDA

The EDA provided a overview of the company operational and customer performance. The Executive Dashboard (see Figure 1) highlights several critical indicators that define the current state of the business.

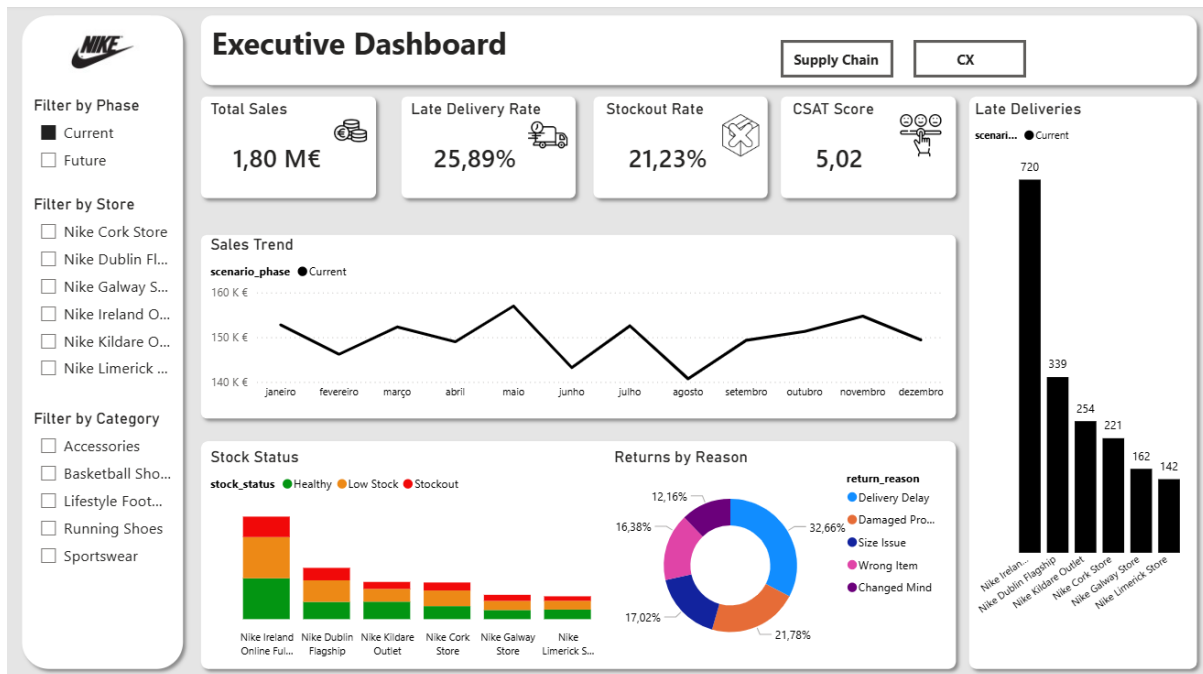


Figure 1 - Executive Dashboard

According to the data, the total revenue remains in a good way but operational efficiencies are bad. The Late Delivery Rate is at approximately 26%, which means that one out of five orders are delivered later than expected. This is a significant issue because timely delivery is a key component of customer satisfaction. Similarly, the stockout rate of 21%, suggests that inventory management processes are not aligned with the demand, and this also causes late deliveries because of product unavailability. The impact of the current situation on customers is also negative, according to the Customer Experience dashboard (see Figure 2) the scores are low with a Customer Satisfaction (CSAT) of 5,02, the CSAT is measured on a scale of 10, and indicates a moderate CSAT level. Currently, customers wait on average, 56 hours for customer support, and this shows that the company is not fast and therefore, customer support is not helpful.

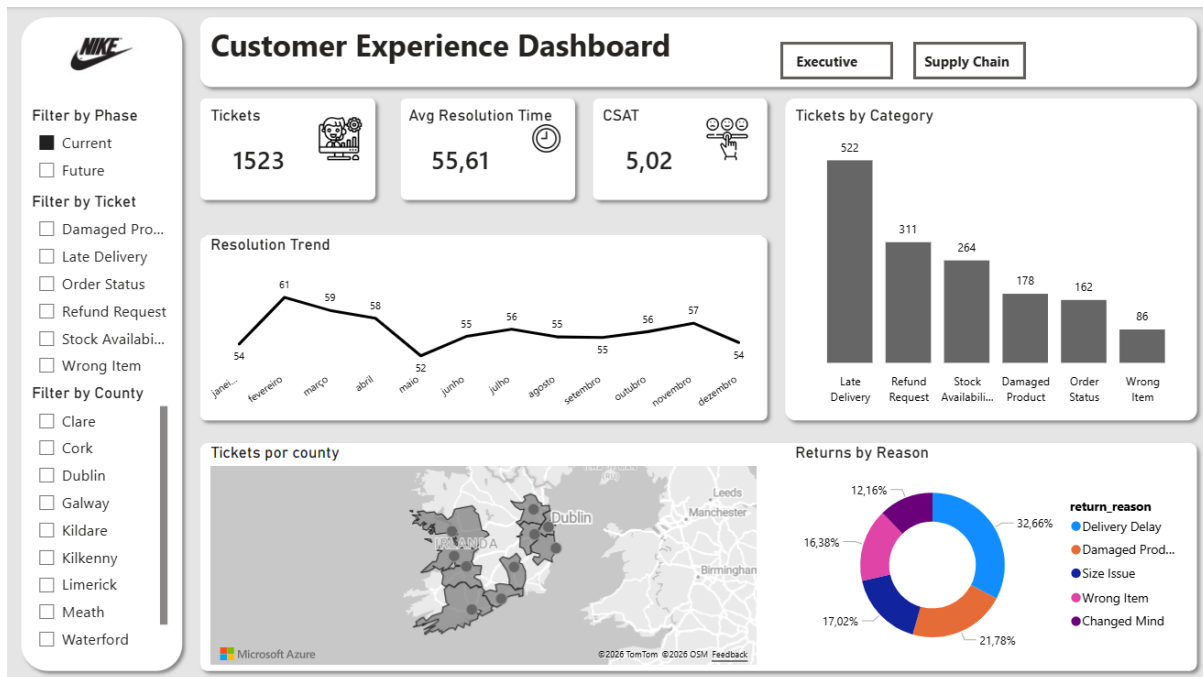


Figure 2 - Customer Experience Dashboard

A deeper analysis of customer tickets shows that late deliveries are the most frequent complaint, this shows that operational inefficiencies within the supply chain are directly impacting the customer experience.

The Supply Chain dashboard (see figure 3) provides insights into the root of the problem, the supplier performance scores is around 71,43 out of 100 in average, some suppliers perform consistently well while others introduce delays that disrupt the entire delivery process, mostly suppliers from Asia.

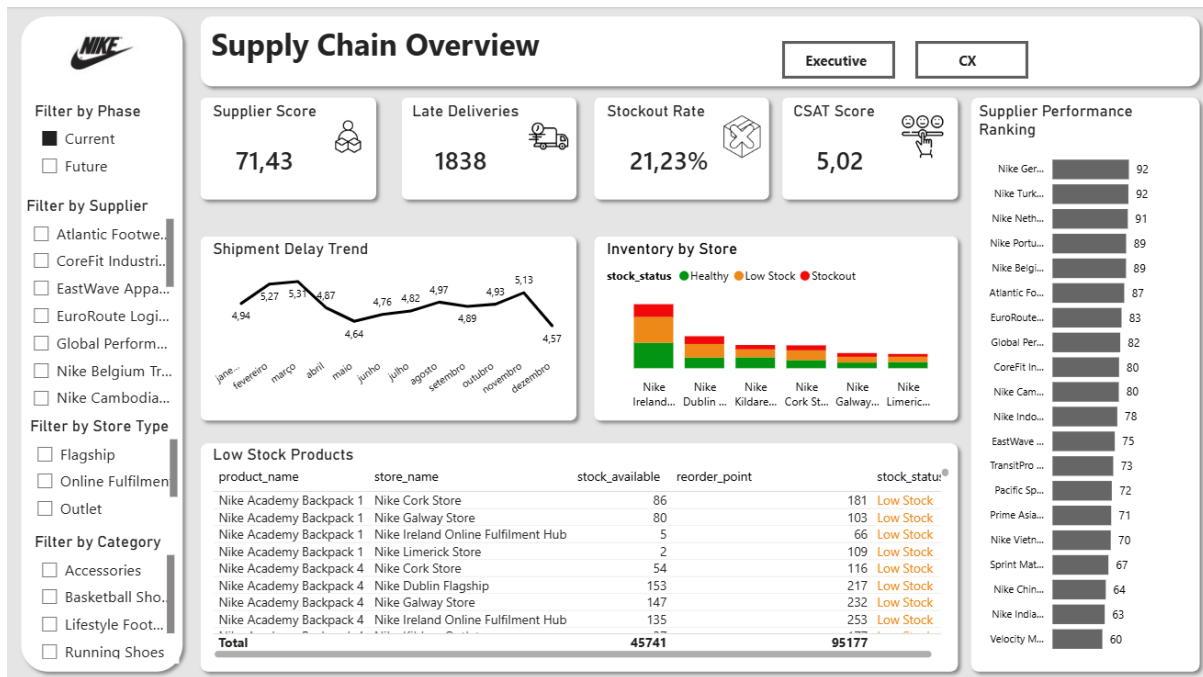


Figure 3 - Supply Chain Dashboard

For the EDA we proposed the following questions:

- Which stores have the highest stock pressure?
- Which products are the most frequently out of stock and low stock?
- Which suppliers have the worst delivery performance?
- Which return reasons are most linked to operational failures?
- How do the current and future scenarios compare across key KPIs?

II. SWOT & Gap analysis

Table 1 – SWOT Analysis

Strengths	Weaknesses	Opportunities	Threats
Strong global brand presence; Loyal customer base in Ireland; Diversified sales channels including retail, outlet and online; Strong demand for core product lines.	Poor visibility into shipment and supplier performance; Delayed deliveries affecting customer trust; Inventory imbalance between stores and online fulfilment.	Implement dashboard for operational visibility; Use of CRM to improve ticket and customer issue management; Improve supplier monitoring.	Customer churn due to poor experience; Loss of sales caused by products out of stock; Competitive pressure from Adidas, JD Sports, and other sportswear retailers.

Gap Current State

The supply chain and customer support processes have limited visibility with the current internal operations, shipments are consistently delayed, there is a lack of available stock, and customers are receiving delayed responses, and it reduces the operational performance.

Desire State

Nike Ireland intends to improve its customer engagement operations and become more customer centric, and data driven through increased visibility with suppliers, improved management of stock, quicker resolution of support and service performance and improved service support.

Identified Gap

The gap between the current and future state can be described as the inability to operationalise data insights, right now, data is simply collected and is not being used for decision making, and for the future state, this gap is closed by implementing an automated workflow that ensure that insights lead to immediate and consistent actions.

Action Points

- Optimise and centralise operational and customer data;
- Use of a BI software (Power BI) to implement dashboards;
- Use of a CRM software (Hubspot) to optimise ticketing and customer workflow;
- Monitor regularly and evaluate performance;
- Compare current vs future KPIs to evaluate operational recovery.

b) System design

To achieve actionable insights, the proposed system addresses the existing operational gap by combining Business Intelligence and CRM processes. The system follows a layered architecture consisting of a data layer, an analytics layer, and an operational layer. The data layer consolidates information from multiple sources, including orders, inventory, shipments and customer tickets and this data is structured into a relational model that supports efficient analysis. The analytics layer, which is implemented using Power BI, is responsible for transforming raw data into actionable insights, through dashboards with KPIs such as late delivery rates, stockout rates, and customer satisfaction rate. The operational layer is implemented using Hubspot CRM, unlike traditional reporting systems that generate static insights, the operational layer generates automated actions defined by the system conditions.



Figure 4 - System Architecture Design

c) Database design

The model consists of fact tables such as orders, shipments, inventory, customer tickets and returns, these tables capture transactional data and link to dimension tables that provides contextual information such as customer details, product attributes, supplier information and store locations.

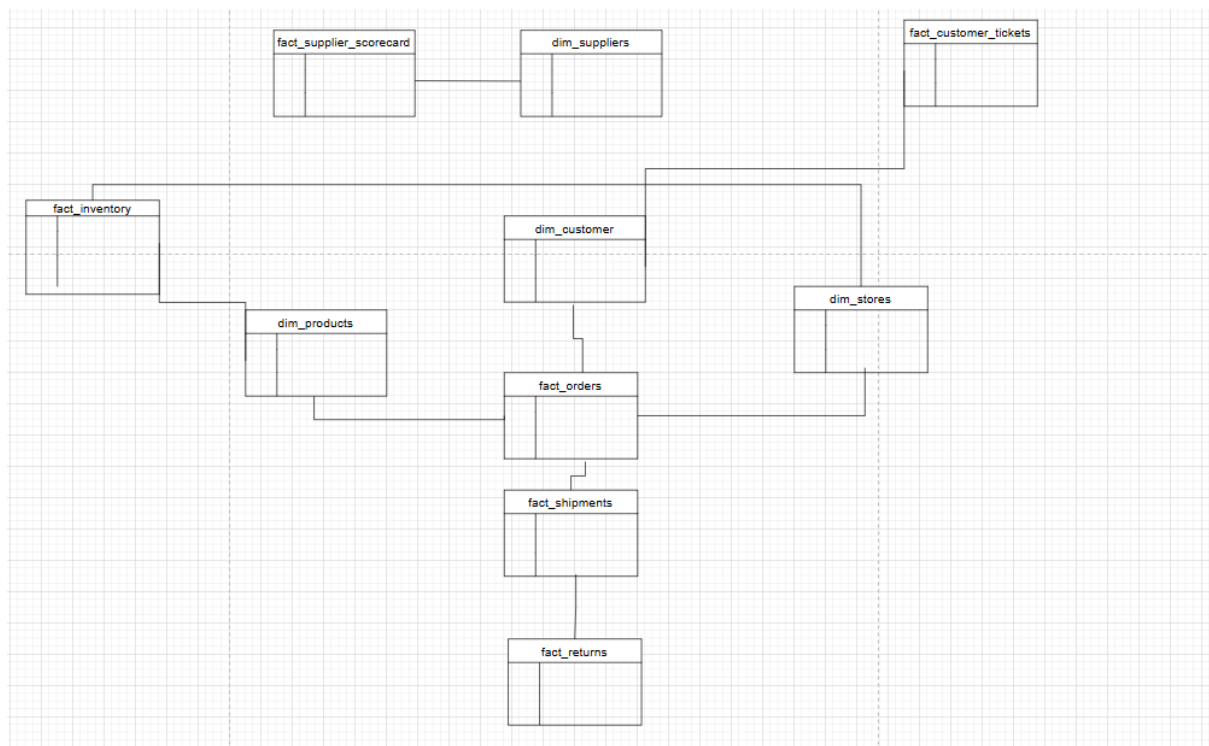


Figure 5 – Database design

Data Dictionary – Dimension Tables

Table 2 - Dim_customers

Field	Type	Description
Customer_id	Integer	Unique identifier for each customer

Customer_name	Text	Full name of the customer
County	Text	Customer's county
City	Text	Customer's city
Loyalty_tier	Text	Loyalty segment
Channel_preference	Text	Preferred channel
Customer_type	Text	Type of customer
Signup_date	Date	Date customer registered

Table 3 - Dim_products

Field	Type	Description
Product_id	Integer	Unique product identifier
SKU	Text	Stock Keeping Unit
Product_name	Text	Name of product
Category	Text	Product category
Sub_category	Text	Product subcategory
Launch_year	Text	Year product launched
Standard_cost	Decimal	Cost price
Unit_price	Decimal	Selling price
Is_core_product	Text	Indicates key product
Supplier_id	Integer	Foreign Key

Table 4 - Dim_stores

Field	Type	Description
Store_id	Integer	Unique store identifier
Store_name	Text	Store_name
City	Text	Store city
County	Text	Store county
Store_type	Text	Store type
Opening_date	Date	Opening date
Capacity_units	Integer	Storage capacity

Table 5 - Dim_suppliers

Field	Type	Description
Supplier_id	Integer	Unique supplier ID
Supplier_name	Text	Supplier name
Supplier_country	Text	Country of origin
Supplier_region	Text	Region
Supplier_type	Text	Supplier type
Contract_start_date	Date	Contract start
Lead_time_days	Integer	Average lead time
Reliability_score	Decimal	Supplier reliability
Quality_score	Decimal	Product quality rating

On_time_delivery_rate	Decimal	% on time deliveries
Risk_level	Text	Risk classification

Fact Tables

Table 6 - Fact_orders

Field	Type	Description
Order_id	Integer	Unique order ID
Order_date	Date	Date of order
Customer_id	Integer	Foreign key
Product_id	Integer	Foreign Key
Store_id	Integer	Foreign Key
Quantity	Integer	Units sold
Unit_price	Decimal	Price per unit
Discount_amount	Decimal	Discount applied
Gross_sales	Decimal	Total before discount
Net_sales	Decimal	Total after discount
Delivery_delay_days	Integer	Days delayed
Delivery_status	Text	Delivery status
Sales_channel	Text	Sales channel
Scenario_phase	Text	Scenario

Table 7 - Fact_shipments

Field	Type	Description
Shipment_id	Integer	Shipment ID
Supplier_id	Integer	Foreign key
Product_id	Integer	Foreign key
Store_id	Integer	Foreign key
Planned_ship_date	Date	Planned shipment date
Planned_arrival_date	Date	Expected arrival
Actual_arrival_date	Date	Actual arrival
Shipment_quantity	Integer	Units shipped
Shipment_cost	Decimal	Cost of shipment
Delay_days	Integer	Shipment delay
Shipment_status	Text	Shipment status
Scenario_phase	Text	Scenario

Table 8 - Fact_inventory

Field	Type	Description
Snapshot_id	Integer	Inventory snapshot ID
Snapshot_date	Date	Date of snapshot
Product_id	Integer	Foreign Key

Store_id	Integer	Foreign Key
Stock_on_hand	Integer	Total stock
Stock_reserved	Integer	Reserved stock
Stock_available	Integer	Available stock
Reorder_point	Integer	Reorder threshold
Days_of_cover	Decimal	Days stock can last
Stock_status	Text	Stock status
Scenario_phase	Text	Scenario

Table 9 - Fact_customer_tickets

Field	Type	Description
Ticket_id	Integer	Ticket ID
Customer_id	Integer	Foreign Key
Order_id	Integer	Foreign Key
Ticket_created_date	Date	Ticket creation
Ticket_category	Text	Issue type
Priority	Text	Priority
Resolution_hours	Decimal	Time to resolve
Csat_score	Decimal	Satisfaction score
Scenario_phase	Text	Scenario

Table 10 - Fact_returns

Field	Type	Description
Return_id	Integer	Return ID
Order_id	Integer	Foreign Key
Customer_id	Integer	Foreign Key
Product_id	Integer	Foreign Key
Return_date	Date	Date of return
Return_reason	Text	Reason
Refund_amount	Decimal	Refund value
Replacement_requested	Text	Replacement required
Scenario_phase	Text	Scenario

Table 11 – fact_supplier_scorecard

Field	Type	Description
Scorecard_id	Integer	Scorecard ID
Supplier_id	Integer	Foreign Key
Scorecard_month	Date	Month
On_time_delivery_rate	Decimal	% on time
Avg_lead_time_days	Decimal	Average lead time
Defect_rate	Decimal	Defect %
Supplier_score	Decimal	Overall performance

Scenario_phase	Text	Scenario
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d) Mock up data & data structures

Unfortunately, real data are not available, and to support the implementation of the system, mock datasets were generated on Mockaroo, data preprocessing involved the creation of calculated columns, measures, standardising categorical variables, and deriving KPIs such as late delivery and stockout rates.

2. Project implementation

a) Development process

The development process followed a structured process, we started with data exploration and progressing through modelling, then, the dashboard development and CRM integration. Initially, the focus was on building a unique dashboard to visualise KPIs regarding the supply chain, however, as the project evolved, it became clear that just one dashboard will not be sufficient to address the current challenge, we decided to design and implement more two dashboards, an executive and a customer satisfaction dashboard, and also integrate CRM into the current phase. For the CRM as the account is limited, we imported a few contacts, created a good number of tickets and created a workflow.

b) System implementation

For the CRM implementation we used HubSpot, where the ticket management system was configured to capture and process customer issues, a workflow was developed to automate the handling of late delivery cases, specifically when a ticket is categorised as Late Delivery, it automatically updates its status to Under Investigation and assigns it to a customer service agent that we named Daniel_Hen. This automation ensures that critical issues are prioritised and handled immediately, reducing response times and improving service consistency.

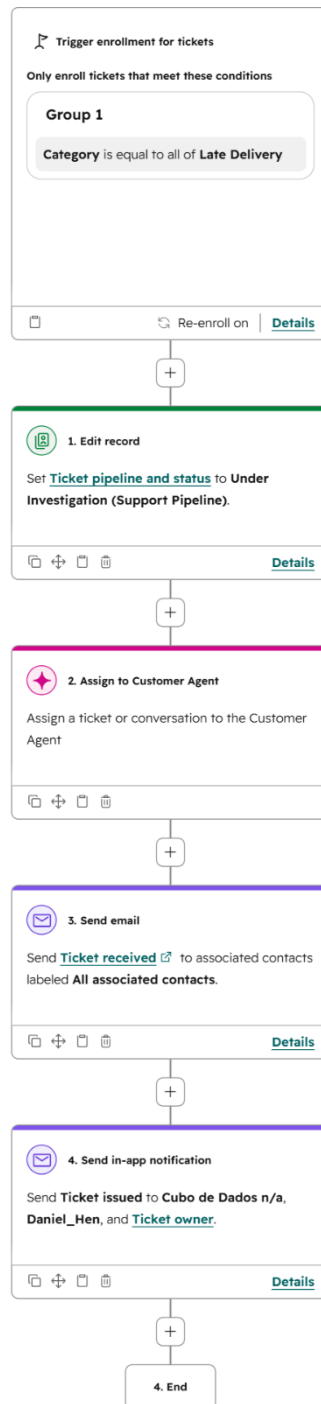


Figure 6 – Customer Support Automation Workflow

The dashboards were created using Power BI, the modelling phase consisted on creating relationship between tables as follows below (see figure 7), and also some key measures such as CSAT, late deliveries rate, and stockout rate, and many others, because they are important indicators to improve the business.

Figure 7 – Modelling View

c) Reports and dashboards

We developed three dashboards as part of the solution, each serving a distinct purpose, the executive dashboard provides a high-level overview of the business performance, focusing on key metrics such as total sales, delivery performance, customer satisfaction and also filters to the current and future scenarios. The supply chain dashboard offers insights into supplier performance, inventory levels and shipment delays, and the Customer Experience Dashboard focuses on resolution trends, ticket analysis and satisfaction metrics. All three dashboards can be filtered into current and future phase, where we can see improvements, and they provide a comprehensive view of Nike's performance giving them the possibility to take informed decision making at both strategic and operational levels.

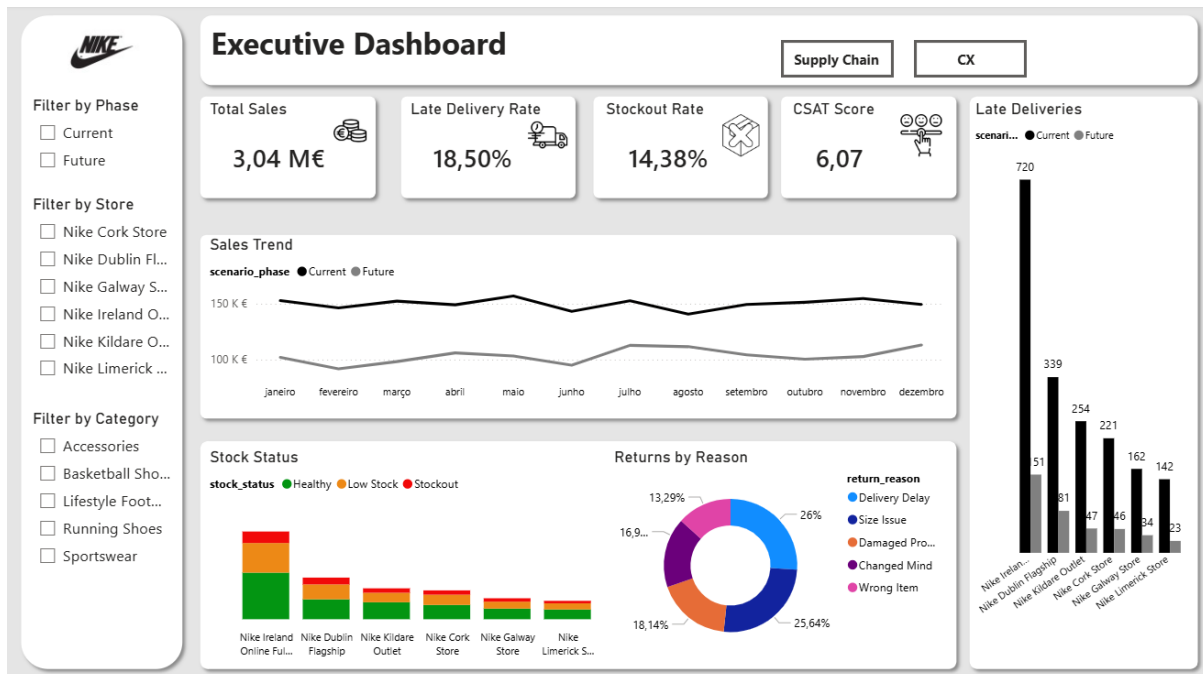


Figure 8 – Executive Dashboard

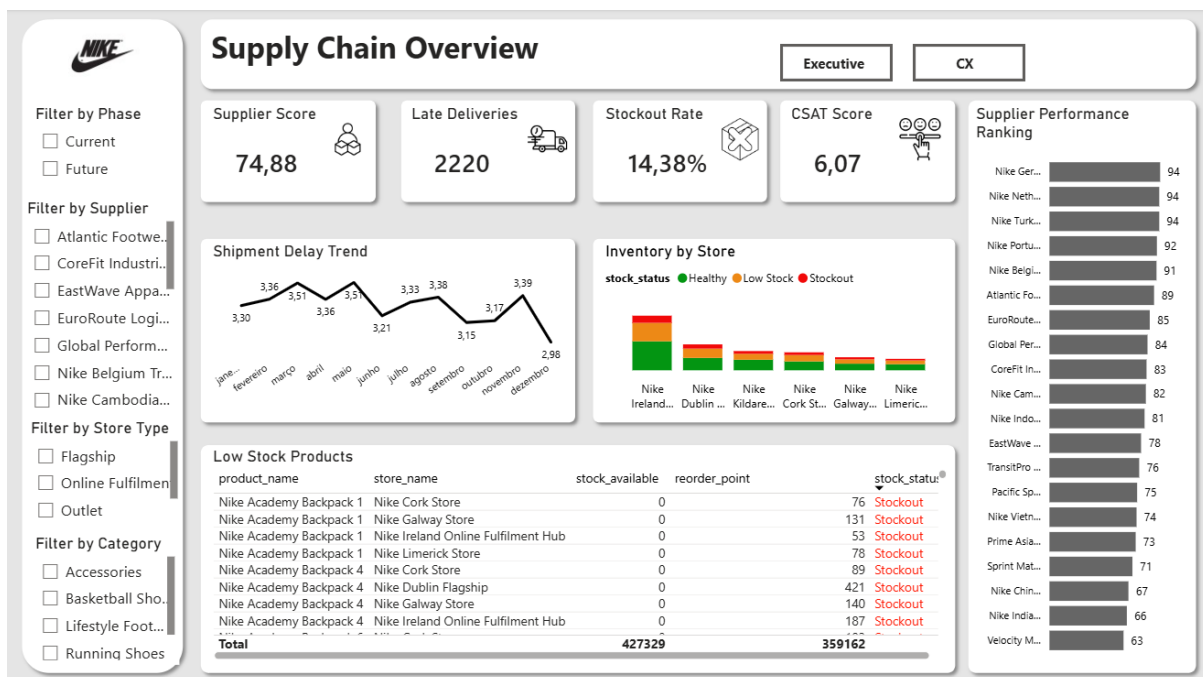


Figure 9 – Supply Chain Dashboard

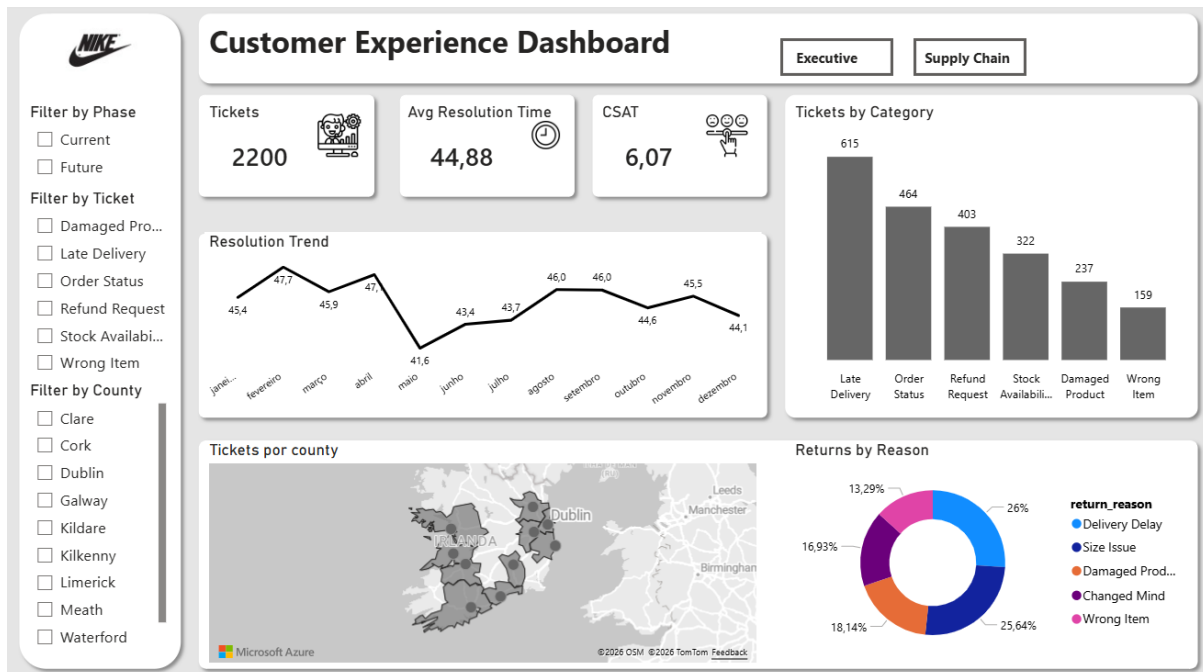


Figure 10 – Customer Experience Dashboard

d) Initial response

The implementation is expected to result in significant improvements across KPIs, by automating late deliveries handling, Nike Ireland can reduce response times and ensure more consistent service delivery. Preliminary analysis suggests that CSAT scores could improve significantly, potentially increasing from 5,02 to 8,44. Additionally, the reduction in manual processes is expected to improve operational efficiency and reduce the workload on customer service teams with the help of an automated agent.

e) Conclusions & future work

By turning Nike Ireland into a data driven company, combining BI and CRM, the company could transform its operations into a proactive system. For future work we will focus on improving the system with predictive analysis, in order to help the company to anticipate and prevent issues before they occur.

References

[1] T. H. Davenport, "Competing on Analytics," *Harvard Business Review*, Jan. 2006.
<https://hbr.org/2006/01/competing-on-analytics>

Author contributions (for teamwork submissions)

Author	Sections written (in %)	Sections reviewed (in %)
Henrique Carlos da Silva Ricardo	EDA (10%) System Design (10%) Data Dictionary (10%) Development Process (5%) Reports and dashboards (10%) Conclusion and future work (5%)	Background information and business goals (10%) SWOT and gap analysis (10%) Database design (5%) Mock up data and data structures (5%) System implementation (10%) Initial Response (10%)
Daniel Tijesu Ohifeme	Background information and business goals (10%) SWOT and gap analysis (10%) Database design (5%) Mock up data and data structures (5%) System implementation (10%) Initial Response (10%)	EDA (10%) System Design (10%) Data Dictionary (10%) Development Process (5%) Reports and dashboards (10%) Conclusion and future work (5%)